DAY 26 COMPLETED

RetailPulse Monitoring Report

System Monitoring & Visualization with Prometheus and Grafana

📌 Executive Summary

This report documents the completion of **Day 26: Monitoring setup with Prometheus and Grafana** for the **RetailPulse** application. The setup establishes real-time telemetry, metric scraping, and dashboard visualization for monitoring system performance and ML model execution metrics.

1. Architecture & Metric Configuration

Component	Target Metric / Endpoint	Purpose
Prometheus	`http://localhost:9090`	Scrapes and stores time-series metric data from the application target.
Grafana	`http://localhost:3000`	Visualizes time-series data using customized dashboard panels.
Metric 1	`model_execution_time_seconds_sum`	Tracks cumulative execution time (in seconds) for machine learning model runs.
Metric 2	`app_visits_total` / Counter	Monitors overall application request frequency and traffic visits.

2. Dashboard Setup (RetailPulse Monitoring)

Panel Details

- **Panel 1:** Total Application Visits (Counter visualization).
- **Panel 2:** `Model Execution Time` (Time Series visualization for model latency/duration).
- **Refresh Rate:** Configured for `5s` auto-refresh to deliver real-time metric updates.

3. Deployment Verification

Status: ✅ Fully Operational
Prometheus is actively scraping metrics without parsing errors, and Grafana is rendering live time-series graphs on the `RetailPulse Monitoring` dashboard.